

Comparing speaker-embedding models

Sprint 1 · Speaker identification · Comparing 3 pre-trained speaker-embedding models

Smart Subtitle Glasses for hearing-impaired children · Mechatronics FYP, NUST · Ayesha & Sabeeh · September 2026

Background

The glasses must show **who** is speaking. Each voice clip is turned into a **voice fingerprint** (embedding). A new clip is compared with each enrolled person using **cosine similarity**. If the best score passes a **threshold**, it gets that person's name; otherwise it is labelled **UNKNOWN**. Models are usually ranked by **EER** (equal error rate, lower is better).

Method

- **Data:** 2 enrolled speakers (5–30 s clips), 10 known test clips, 4 unknown-speaker clips; clean speech.
- **Same pipeline** for every model: embed → score every clip vs every speaker → decide.
- **Fair threshold:** set on odd-numbered clips, tested on even-numbered clips.
- **d'** = how far apart genuine and impostor scores are (comparable across models).

The three models

WeSpeaker ResNet34-LM

CNN · supervised on VoxCeleb2
6.6M params · 256-d embedding
Published EER **≈0.8%**
ONNX / ARM-ready

TitaNet-Large (NVIDIA)

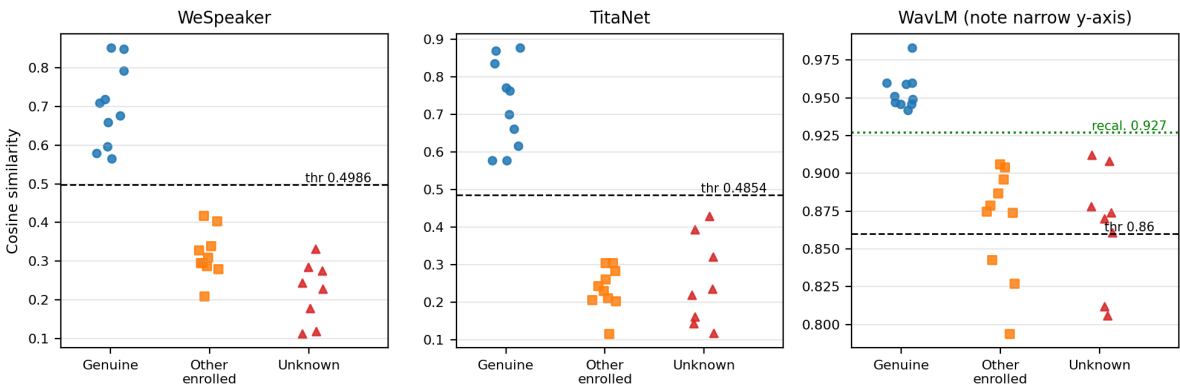
1-D separable CNN · large English mix
≈23M params · 192-d (TitaNet-S **≈6M**)
Published EER **≈0.68%**
ONNX / ARM-ready

WavLM Base+ SV (Microsoft)

Self-supervised Transformer
≈95M params · raw waveform input
EER **≈4%** (closest published setup)
No edge export found

Results on our recordings

	WeSpeaker	TitaNet	WavLM (default)	WavLM (recalibrated)
Threshold	0.4986	0.4854	0.86	0.927
Known speakers correct	10/10	10/10	10/10	10/10
Strangers rejected	4/4	4/4	0/4	4/4
Safety gap (genuine – impostor)	0.147	0.147	0.030	0.030
Separation d'	4.49	4.72	3.22	3.22
Stranger closest to threshold	0.17 below	0.06 below	all above	0.015 below



Each dot is one score. Blue = same speaker, orange = the other enrolled speaker, red = stranger. Dashed line = threshold. WeSpeaker and TitaNet leave a clear gap; WavLM squeezes every score between 0.79 and 0.98.

What it means

Key findings

- **WeSpeaker and TitaNet performed similarly.** Both identified every known clip and rejected every stranger. TitaNet separated the two enrolled speakers more; WeSpeaker kept strangers further below its threshold.
- **WavLM** accepted all 4 strangers at its default threshold (0.86). Recalibrated (0.927) it was correct, with a much smaller safety gap.
- **Enrollment length matters:** 5–10 s enrollment caused some false rejects in WeSpeaker and TitaNet; ≥ 15 s had none.

Observations & next steps

- All errors came from **one speaker** in both WeSpeaker and TitaNet, pointing to recording quality rather than the model.
- Published results rank the models the same way: WeSpeaker and TitaNet ≈ 0.7 – 0.8% EER, WavLM setup $\approx 4\%$.
- **Next:** re-record enrollment (≥ 15 s, two sessions) · add more strangers · re-run all models on identical data · noise and Urdu tests · measure latency on Raspberry Pi 5.

Caveats: pilot only — 2 speakers, 10 known + 4 unknown clips (10/10 has a 95% interval of 72–100%). Models did not use identical enrollment data; WavLM ran via a separate script and was recalibrated afterwards. Published EERs come from different papers and setups. No latency has been measured on the Pi yet. Full report and code: github.com/a-malik00/AURIS